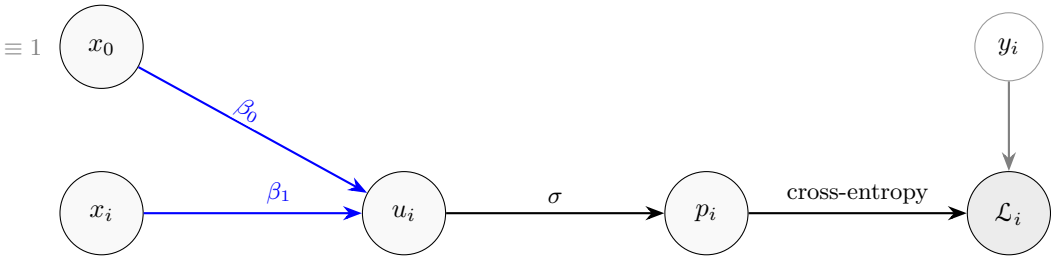
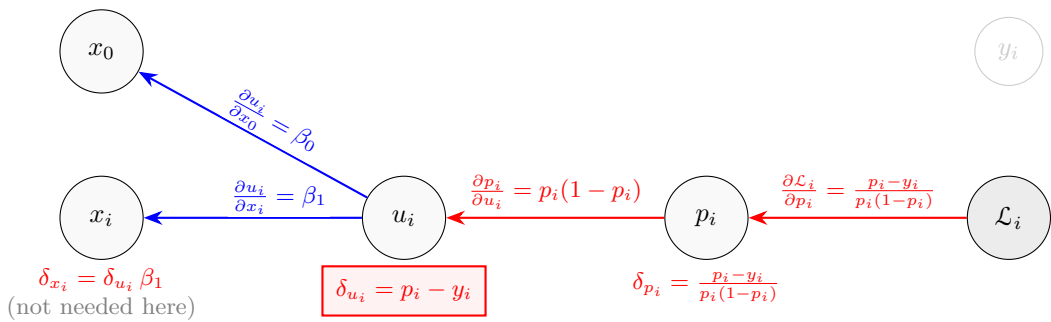
forward pass: values



backward pass: local derivatives



$$\frac{\partial \mathcal{L}_i}{\partial \beta_1} = \delta_{u_i}\, x_i$$

$$\frac{\partial \mathcal{L}_i}{\partial \beta_0} = \delta_{u_i}\, x_0 = \delta_{u_i}$$